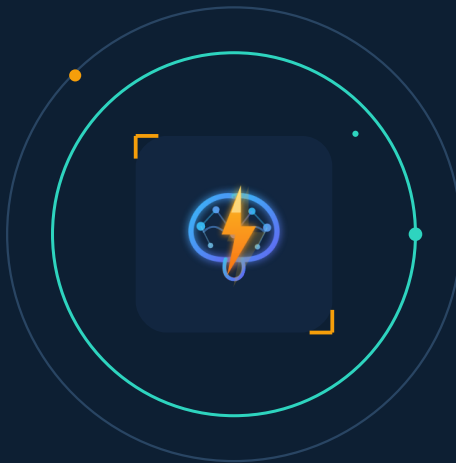




B R A I N S P A R K

HORIZON

Automotive Technology Foresight — Live Run Report

10 SYSTEMS · ICE / MHEV / PHEV / BEV · ENGINE-COMPUTED · TWO-CHAIR AUDITED

- | | | | |
|----|---------------------------|----|-------------------------|
| 01 | BEV HV Battery Pack | 06 | Wiring Harness & E/E |
| 02 | Electric Drive Unit (EDU) | 07 | ADAS & Perception |
| 03 | Braking System | 08 | Thermal & Heat Pump |
| 04 | Suspension | 09 | ICE / Hybrid Powertrain |
| 05 | Body-in-White (BIW) | 10 | Cockpit & Displays |

Register v2026 · 160 technologies · Avinash Bhosale · Cost Engineering · 2026-07-28

Every figure in this report was computed by deterministic engines — Bass diffusion, Wright's law, S-curve positioning — over a curated, evidence-locked register. The AI narrates; it never invents a number.

How to read this report

INSTRUMENT LEGEND

 HORIZON LANES	H1 Now–2028: adopt/quote now · H2 2029–2032: plan the transition · H3 2033+: track, don't commit. Only in-force/adopted law may pull a lane forward.
 CONFIDENCE	COMMITTED = regulation or named production programme · PROBABLE = TRL >= 7 · SPECULATIVE = earlier. Colour-coded on every card.
 MOMENTUM 0-100	Maturity + adoption + cost trajectory + drivers + regulation + production evidence. Deterministic composite.
 CROSSING YEARS	When modelled adoption crosses 50% of its segment, with a q+/-25% uncertainty band: ~2031 (2030-2033). Niche techs honestly "never cross".
 ADOPTION CURVES	Bass diffusion (p=0.03, q=0.38) with per-technology segment ceilings, seeded from curated current shares.
 REGULATORY CHIPS	Teal IN FORCE · gold ADOPTED · grey PROPOSED · red UNDER REVISION. A proposal is context, never a commitment.

Provenance & audit

Live run executed against register v2026 (160 technologies; evidence rules, adoption ceilings and legal-status gates CI-enforced). This report was itself audited before release: read as Automotive Chief Engineer for domain authenticity and as AI Orchestrator for pipeline defects — findings and fixes are recorded on the closing page.

01 / 10

BEV HV Battery Pack

The fastest-moving cost centre in the vehicle

query:"BEV HV battery pack" powertrain:BEV · 27 TECHNOLOGIES MAPPED

HORIZON DISTRIBUTION



800V pack architecture

COMMITTED

TRL 9 · GROWTH · ADOPTION 15% · H1

MOMENTUM 65

Halves current for equal power: thinner cables, faster charging; SiC availability made it mainstream.
Porsche Taycan (2019); Hyundai E-GMP (2021)

Full-domain 1000V architecture + megawatt...

COMMITTED

TRL 8 · TAKEOFF · ADOPTION 1% · H1

MOMENTUM 54

BYD Super e-Platform (2025) runs the whole vehicle at 1000V with 1MW flash charging; second-gen 2.1MW stations rolling out 2026.
BYD Han L / Tang L (2025)

LFP chemistry in C/D-segment packs

COMMITTED

TRL 9 · GROWTH · ADOPTION 40% · H1

MOMENTUM 73

Cobalt-free, ~20-30% cheaper per kWh; energy density gap closing via pack integration.
Tesla Model 3 SR (2021), BYD fleet-wide

Cell-to-pack / cell-to-body integration

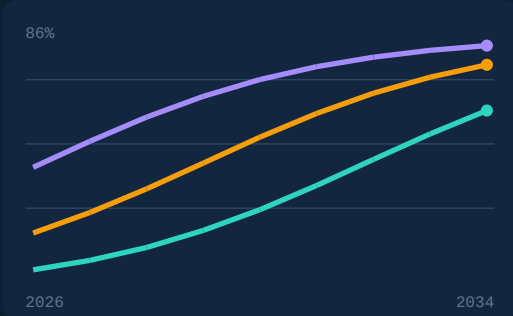
COMMITTED

TRL 9 · GROWTH · ADOPTION 25% · H1

MOMENTUM 69

Deletes module hardware: -15-20% pack cost, +10-15% volumetric density; CTB makes pack a body member.
BYD Blade CTP (2020); Tesla structural 4680 (2022)

MODELLED ADOPTION PATHS -> 2034



- 800V pack architecture
- Full-domain 1000V architecture + megawatt charging
- LFP chemistry in C/D-segment packs

CROSSING CALLS (50% OF SEGMENT)

800V pack architecture	~2030 ('29-'31)
Full-domain 1000V architecture + megawatt charging	~2033 ('32-'34)
LFP chemistry in C/D-segment packs	~2027 ('27-'27)
Cell-to-pack / cell-to-body integration	~2029 ('28-'29)

REGULATORY ANCHORS IN PLAY

- EU Battery Regulation — digital... ADOPTED
- EU Battery Regulation — recycle... ADOPTED
- EU CBAM definitive regime · 2026 IN-FORCE
- China GB 38031-2025 battery safety... IN-FORCE

ENGINE READ

Lead technology: 800V pack architecture — momentum 65/100, COMMITTED, modelled to cross 50% ~2030. 21 of 27 technologies are in the act-now window; 4 regulations anchor this landscape. Cost index of the leader at +5y: 0.77.

ALSO ON THE RADAR

- Thermal-propagation barriers (mica/aerogel/intumescent) — TRL 9, momentum 69, H1
- Pyro-fuses & integrated battery disconnect units (BDU/BJB) — TRL 9, momentum 69, H1

02 / 10

Electric Drive Unit (EDU)

Motor, inverter, reducer — the electric heart

commodity:EDU · 14 TECHNOLOGIES MAPPED

HORIZON DISTRIBUTION



SiC power stage as volume default

COMMITTED

TRL 9 · GROWTH · ADOPTION 35% · H1

MOMENTUM 79

800V + efficiency gains; 200mm wafer ramp collapses the premium — IGBT retreats to entry 400V.

Tesla Model 3 (2017), now industry-wide

8-layer / X-pin / W-pin windings

COMMITTED

TRL 9 · GROWTH · ADOPTION 20% · H1

MOMENTUM 67

Higher slot fill -> -10-20% copper; continuous-wave variants cut weld count massively.

XPeng G6 (2023); NIO ET9 (2024)

Integrated 6-in-1 / 8-in-1 drive units

COMMITTED

TRL 9 · GROWTH · ADOPTION 15% · H1

MOMENTUM 65

Deletes housings, cables, coolant loops: -20-30% e-drive system cost at volume.

BYD e-platform 3.0 (2021)

Ultra-thin electrical steel & bonded laminatio...

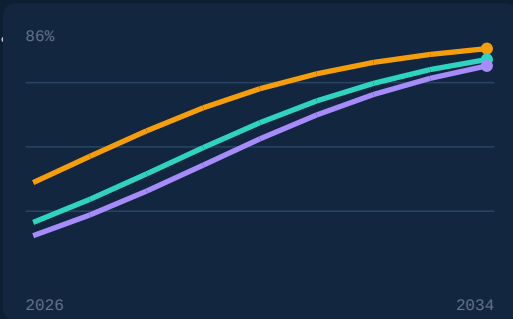
COMMITTED

TRL 9 · TAKEOFF · ADOPTION 12% · H1

MOMENTUM 64

The gauge ladder is the prediction: 0.35 -> 0.25/0.20 (Western premium EDUs) -> 0.15mm already in Chinese production (Xiaomi V8s EVO, 960 MPa high-strength Xiaomi V8s 0.2mm / V8s EVO 0.15mm (2023-25); Lucid & BMW Gen5-class N025...

MODELLED ADOPTION PATHS -> 2034



- SiC power stage as volume default
- 8-layer / X-pin / W-pin windings
- Integrated 6-in-1 / 8-in-1 drive units

CROSSING CALLS (50% OF SEGMENT)

SiC power stage as volu...	~2028 ('27-'28)
8-layer / X-pin / W-pin wi...	~2029 ('29-'30)
Integrated 6-in-1 / 8-in-1...	~2030 ('29-'31)
Ultra-thin electrical steel...	~2030 ('30-'31)

REGULATORY ANCHORS IN PLAY

- EU ELV Regulation recast · 2032 ADOPTED

ENGINE READ

Lead technology: SiC power stage as volume default — momentum 79/100, COMMITTED, modelled to cross 50% ~2028. 8 of 14 technologies are in the act-now window; 1 regulation anchor this landscape. Cost index of the leader at +5y: 0.76.

ALSO ON THE RADAR

- Direct oil-spray motor cooling — TRL 9, momentum 63, H1
- Externally-excited synchronous motors (magnet-free) — TRL 9, momentum 62, H1

HORIZON DISTRIBUTION

H1:3

One-box integrated brake units

COMMITTED

TRL 9 · GROWTH · ADOPTION 45% · H1

MOMENTUM 77

Already the regen-blending default; supplier competition compressing cost ~10%/generation.
industry-wide since ~2020

Electro-mechanical (dry) brake-by-wire

COMMITTED

TRL 8 · TAKEOFF · ADOPTION 1% · H1

MOMENTUM 69

Deletes hydraulics entirely; per-wheel software control. Brembo Sensify hits series production 2026, Chinese EMB SOPs 2026, Conti/ZF/Bosch follow 2027-28.
Brembo Sensify SOP (2026); BWI EMB (2026, China)

Hard-coated brake discs (Euro 7)

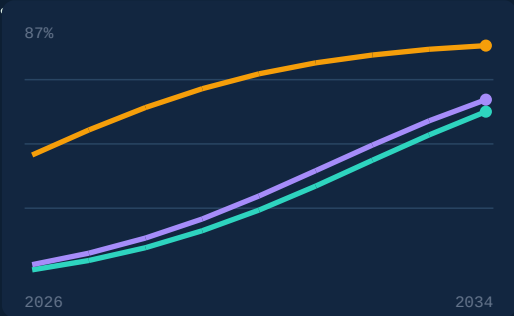
COMMITTED

TRL 9 · TAKEOFF · ADOPTION 3% · H1

MOMENTUM 68

Euro 7 brake-particulate limits (3-7 mg/km, from Nov 2026 type approvals) make HVOF/LMD carbide coatings volume content: -80-90% brake dust.
Porsche PSCB (2017); Brembo Greentell (2025-26)

MODELLED ADOPTION PATHS -> 2034



- One-box integrated brake units
- Electro-mechanical (dry) brake-by-wire
- Hard-coated brake discs (Euro 7)

CROSSING CALLS (50% OF SEGMENT)

One-box integrated brak...	~2027 ('26-'27)
Electro-mechanical (dry)...	~2033 ('32-'34)
Hard-coated brake discs...	~2032 ('31-'33)

REGULATORY ANCHORS IN PLAY

- Euro 7 (M1) · 2027
- ADOPTED

ENGINE READ

Lead technology: One-box integrated brake units — momentum 77/100, COMMITTED, modelled to cross 50% ~2027. 3 of 3 technologies are in the act-now window; 1 regulation anchor this landscape. Cost index of the leader at +5y: 0.90.

Suspension

The off-road & luxury battleground

query:"suspension" · 7 TECHNOLOGIES MAPPED

HORIZON DISTRIBUTION



Software-defined semi-active damping

COMMITTED

TRL 9 · TAKEOFF · ADOPTION 12% · H1

MOMENTUM 70

CDC-class hardware at ~40% of old price via Chinese volume; becoming a mid-trim checkbox.

Chinese CDC-class SOPs fleet-wide (2023-25)

Camera/road-preview predictive suspension

COMMITTED

TRL 9 · TAKEOFF · ADOPTION 3% · H1

MOMENTUM 60

Forward camera + nav data prime the dampers before the bump arrives; ClearMotion in the NIO ET9 (2025) starts the cost collapse of what was S-class-only.

Magic Body Control (2013); Cullinan Flagbearer; NIO ET9 ClearMotion (202...

48V active anti-roll (eARS) trickle-down

COMMITTED

TRL 9 · TAKEOFF · ADOPTION 3% · H1

MOMENTUM 58

48V electromechanical roll control leaves the hydraulic pump behind; Chinese D-segment adoption compresses cost like air suspension did.

Bentley Bentayga 48V eARS (2016) -> NIO/Li Auto

Air suspension cost collapse (China)

PROBABLE

TRL 9 · TAKEOFF · ADOPTION 8% · H1

MOMENTUM 58

Chinese supply chain halved system cost; standard on mid-premium Chinese BEVs, pressuring EU premium pricing.

Vibracoustic · Baolong · NIO · Li Auto

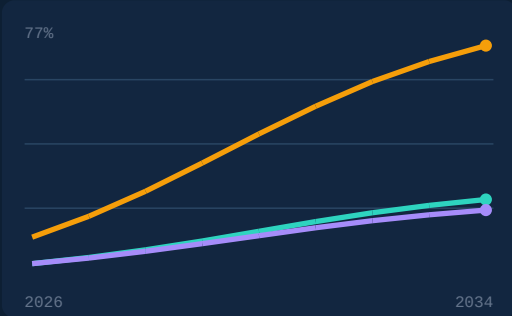
ENGINE READ

Lead technology: Software-defined semi-active damping — momentum 70/100, COMMITTED, modelled to cross 50% ~2030. 6 of 7 technologies are in the act-now window. Cost index of the leader at +5y: 0.57.

ALSO ON THE RADAR

- Hydraulic cross-linked suspension (6D / DiSus-P class) — TRL 9, momentum 57, H1
- GFRP leaf/coil springs — TRL 8, momentum 54, H1

MODELLED ADOPTION PATHS -> 2034



- Software-defined semi-active damping
- Camera/road-preview predictive suspension
- 48V active anti-roll (eARS) trickle-down

CROSSING CALLS (50% OF SEGMENT)

Software-defined semi-a...	~2030 ('30-'31)
Camera/road-preview pre...	NOT IN 15Y
48V active anti-roll (eAR...	NOT IN 15Y
Air suspension cost colla...	NOT IN 15Y

HORIZON DISTRIBUTION



Closed-loop & high-recycled aluminium sheet

COMMITTED

TRL 9 · GROWTH · ADOPTION 15% · H1
MOMENTUM 65

Scrap-tolerant alloys let 50-85% recycled content into closures/structures; CBAM sharpens the case.
Novelis · Hydro · JLR REALCAR lineage

Mega/giga-casting front & rear underbody

COMMITTED

TRL 9 · TAKEOFF · ADOPTION 8% · H1
MOMENTUM 60

-20-40% underbody cost at volume; repairability & insurance pushback is the counterforce; adoption still broadening.
Tesla Model Y (2020) -> 10+ OEMs

Aluminium sill/rocker crash extrusions for p...

COMMITTED

TRL 9 · GROWTH · ADOPTION 20% · H1
MOMENTUM 59

Multi-chamber extrusions absorb pole intrusion before it reaches cells — the default BEV rocker answer, now being challenged by 3D roll-formed steel on cost.
BEV skateboard platforms (2017->)

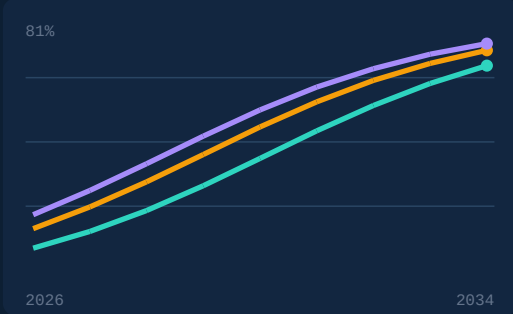
Pack-as-floor structural integration

COMMITTED

TRL 8 · TAKEOFF · ADOPTION 5% · H1
MOMENTUM 58

CTB merges pack lid/floor with body: -10% body-pack mass, stiffness up; service/repair model must adapt.
BYD Seal CTB (2022)

MODELLED ADOPTION PATHS -> 2034



- Closed-loop & high-recycled aluminium sheet
- Mega/giga-casting front & rear underbody
- Aluminium sill/rocker crash extrusions for packs

CROSSING CALLS (50% OF SEGMENT)

Closed-loop & high-recyc...	~2030 ('29-'31)
Mega/giga-casting front...	~2031 ('30-'32)
Aluminium sill/rocker cra...	~2029 ('29-'30)
Pack-as-floor structural i...	~2032 ('31-'33)

REGULATORY ANCHORS IN PLAY

- EU ELV Regulation recast · 2032 ADOPTED
- EU CBAM definitive regime · 2026 IN-FORCE

ENGINE READ

Lead technology: Closed-loop & high-recycled aluminium sheet — momentum 65/100, COMMITTED, modelled to cross 50% ~2030. 9 of 13 technologies are in the act-now window; 2 regulations anchor this landscape. Cost index of the leader at +5y: 0.77.

ALSO ON THE RADAR

- Low-CO2 (H2-DRI/EAF) body steel — TRL 8, momentum 57, H1
- Tailored-tempered PHS with soft zones — TRL 9, momentum 57, H1

HORIZON DISTRIBUTION



Zonal E/E architecture + central compute

COMMITTED

TRL 8 · TAKEOFF · ADOPTION 8% · H1

MOMENTUM 61

-30-40% harness length, ECU count collapses; the enabler of every by-wire and SDV feature on this list.

Tesla (2017->), Rivian Gen2 (2024)

Aluminium + flat-conductor LV wiring

PROBABLE

TRL 8 · TAKEOFF · ADOPTION 5% · H1

MOMENTUM 48

Copper is the harness cost; aluminium for power runs and FFC/FPC for zonal short links cut both /car and kg — automated assembly is the co-benefit.

Leoni · Yazaki · Sumitomo · Aptiv

FPC/CCS flexible cell-contacting systems

COMMITTED

TRL 9 · GROWTH · ADOPTION 35% · H1

MOMENTUM 73

Flex-PCB cell contacting deletes dozens of crimped wires per module: -60-80% sensing-harness mass and automated assembly.

standard in CTP-era packs (2020->)

Multi-gig automotive Ethernet backbone

COMMITTED

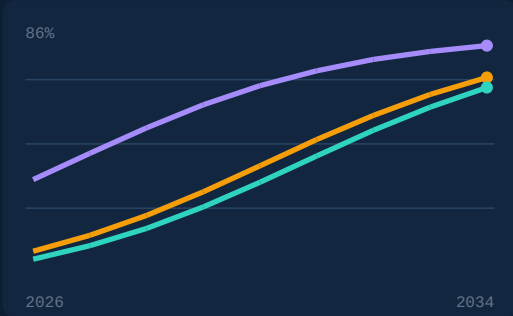
TRL 9 · GROWTH · ADOPTION 20% · H1

MOMENTUM 67

10BASE-T1S to 10G backbone: one network family, deletes gateways; Chinese zonal platforms (NIO Gen2, XPeng) standardise multi-gig backbones (2024->).

BMW X5 camera link (2013) -> backbone in zonal-era EVs

MODELLED ADOPTION PATHS -> 2034



- Zonal E/E architecture + central compute
- Aluminium + flat-conductor LV wiring
- FPC/CCS flexible cell-contacting systems

CROSSING CALLS (50% OF SEGMENT)

Zonal E/E architecture +...	~2031 ('30-'32)
Aluminium + flat-conduct...	~2032 ('31-'33)
FPC/CCS flexible cell-co...	~2028 ('27-'28)
Multi-gig automotive Ethe...	~2029 ('29-'30)

ENGINE READ

Lead technology: Zonal E/E architecture + central compute — momentum 61/100, COMMITTED, modelled to cross 50% ~2031. 4 of 5 technologies are in the act-now window. Cost index of the leader at +5y: 0.71.

ALSO ON THE RADAR

- 48V low-voltage network (12V delete) — TRL 7, momentum 53, H2

HORIZON DISTRIBUTION

H1:10

Onboard AI compute (500-2,000+ TOPS)

COMMITTED

TRL 9 · TAKEOFF · ADOPTION 8% · H1

MOMENTUM 71

End-to-end stacks are compute-hungry: kTOPS-class centralised AI computers become the platform decision, and Chinese in-house SoCs attack the NVIDIA price
NVIDIA Orin fleet-wide (2022->); Thor programmes (2025-26)

End-to-end AI driving stacks (world-model b...

COMMITTED

TRL 8 · TAKEOFF · ADOPTION 5% · H1

MOMENTUM 64

One neural network from pixels to control replaces hand-coded driving logic; China ships it at scale, and the data flywheel becomes the moat — not the sensor set.
Tesla FSD v12 (2024); Huawei ADS 3.0 (2024)

Sub-\$200 lidar

COMMITTED

TRL 8 · TAKEOFF · ADOPTION 6% · H1

MOMENTUM 64

Chinese volume dropped ATX-class lidar under \$200 — the camera-vs-lidar argument is being settled by price.
standard on many Chinese NEVs (2024-25)

AI terrain recognition & adaptive off-road m...

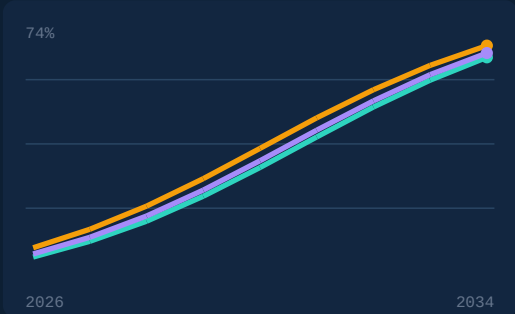
COMMITTED

TRL 9 · TAKEOFF · ADOPTION 10% · H1

MOMENTUM 63

Camera+IMU terrain classification auto-configures diffs, ride height and throttle maps; the dial becomes a fallback as the vehicle reads the surface itself.
Terrain Response 2 Auto (2014->)

MODELLED ADOPTION PATHS -> 2034



- Onboard AI compute (500-2,000+ TOPS)
- End-to-end AI driving stacks (world-model based)
- Sub-\$200 lidar

CROSSING CALLS (50% OF SEGMENT)

Onboard AI compute (500-2,000+ TOPS)	~2031 ('30-'32)
End-to-end AI driving stacks (world-model based)	~2032 ('31-'33)
Sub-\$200 lidar	~2031 ('31-'33)
AI terrain recognition & adaptive off-road mapping	NOT IN 15Y

REGULATORY ANCHORS IN PLAY

- Euro NCAP 2026 protocols · 2026 ADOPTED
- UN R171 DCAS (hands-off L2+)... IN-FORCE

ENGINE READ

Lead technology: Onboard AI compute (500-2,000+ TOPS) — momentum 71/100, COMMITTED, modelled to cross 50% ~2031. 10 of 10 technologies are in the act-now window; 2 regulations anchor this landscape. Cost index of the leader at +5y: 0.52.

ALSO ON THE RADAR

- 4D imaging radar — TRL 9, momentum 61, H1
- Hands-off L2+ / conditional L3 on volume cars — TRL 8, momentum 61, H1

08 / 10

Thermal & Heat Pump

Refrigerant regulation meets integration

query:"heat pump thermal" · 6 TECHNOLOGIES MAPPED

HORIZON DISTRIBUTION

H1:6

Heat pump as standard BEV equipment

COMMITTED

TRL 9 · MAINSTREAM · ADOPTION 55% · H1
MOMENTUM 79

Octovalve-style integrated modules made -20-30% winter range loss recoverable; now a default expectation.
Tesla Model Y (2020) -> mainstream

R744 (CO2) / R290 (propane) refrigerant syst...

COMMITTED

TRL 7 · TAKE OFF · ADOPTION 2%
MOMENTUM 45

F-gas phase-down + PFAS pressure on R1234yf; R744 excels in cold, R290 charge-limits drive compact modules.
Denso · Mahle · VW (R744 ID.)

Thermal-propagation barriers (mica/aerogel/...

COMMITTED

TRL 9 · GROWTH · ADOPTION 30% · H1
MOMENTUM 69

GB 38031 no-fire rule makes cell-to-cell barriers mandatory engineering content worldwide; aerogel sheets are the volume answer.
Aspen Aerogels · Elkem · 3M · Morgan Advanced

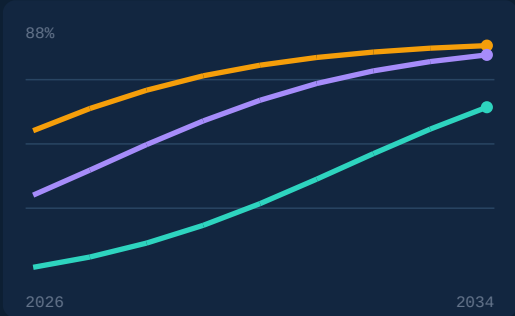
800V SiC e-compressors

COMMITTED

TRL 9 · GROWTH · ADOPTION 20% · H1
MOMENTUM 67

The single biggest HV auxiliary load goes native-800V with SiC inverters onboard: quieter, lighter, and one less voltage domain.
800V platforms (Hyundai E-GMP 2021->)

MODELLED ADOPTION PATHS -> 2034



- Heat pump as standard BEV equipment
- R744 (CO2) / R290 (propane) refrigerant systems
- Thermal-propagation barriers (mica/aerogel/intumesc...

CROSSING CALLS (50% OF SEGMENT)

Heat pump as standard...	PASSED
R744 (CO2) / R290 (prop...	~2032 ('31- '34)
Thermal-propagation barr...	~2028 ('28- '29)
800V SiC e-compressors	~2029 ('29- '30)

REGULATORY ANCHORS IN PLAY

- China GB 38031-2025 battery sa... IN-FORCE
- EU F-gas Regulation phase-dow... IN-FORCE

ENGINE READ

Lead technology: Heat pump as standard BEV equipment — momentum 79/100, COMMITTED, majority adoption already passed. 6 of 6 technologies are in the act-now window; 2 regulations anchor this landscape. Cost index of the leader at +5y: 0.93.

ALSO ON THE RADAR

- Integrated thermal management modules — TRL 9, momentum 65, H1
- Immersion-cooled packs — TRL 6, momentum 39, H1

HORIZON DISTRIBUTION

H1:9

Diesel aftertreatment consolidation (SCR + ...)

COMMITTED

TRL 9 · MAINSTREAM · ADOPTION 90% · H1

MOMENTUM 75

Diesel does not die, it consolidates: close-coupled SCR-on-filter bricks and single-module packaging carry shrinking volumes through Euro 7 — unit costs RISE Euro 6d fleet-wide (2019->)

48V ultra-mild hybrid as ICE floor

COMMITTED

TRL 9 · GROWTH · ADOPTION 30% · H1

MOMENTUM 71

P0/P2 48V becomes the minimum compliance content on every remaining ICE; costs keep compressing.

Audi SQ7 (2016) -> volume rollout across EU OEMs

Electric water pumps + split cooling

COMMITTED

TRL 9 · GROWTH · ADOPTION 30% · H1

MOMENTUM 71

On-demand electric coolant flow and map-controlled thermal management: faster warm-up, 1-2% CO2, and deletes the belt run.

BMW EfficientDynamics (2008->) -> industry-wide

Gasoline particulate filters on every GDI

COMMITTED

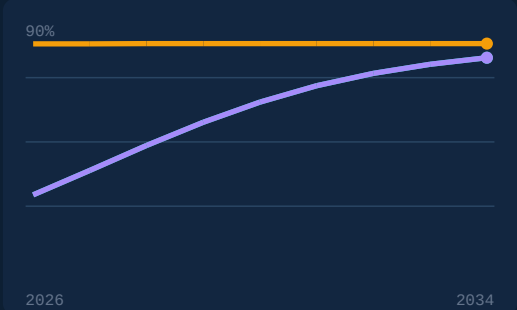
TRL 9 · MAINSTREAM · ADOPTION 70% · H1

MOMENTUM 69

Already near-universal on EU GDI since Euro 6d; Euro 7 tightens PN limits and locks GPF content into every remaining combustion programme.

NGK · Corning · Forvia · Tenneco

MODELLED ADOPTION PATHS -> 2034



■ Diesel aftertreatment consolidation (SCR + DPF + A...)

■ 48V ultra-mild hybrid as ICE floor

■ Electric water pumps + split cooling

CROSSING CALLS (50% OF SEGMENT)

Diesel aftertreatment con... PASSED

48V ultra-mild hybrid as I... ~2028 ('28-'29)

Electric water pumps + s... ~2028 ('28-'29)

Gasoline particulate filter... PASSED

REGULATORY ANCHORS IN PLAY

● Euro 7 (M1) · 2027 ADOPTED

ENGINE READ

Lead technology: Diesel aftertreatment consolidation (SCR + DPF + AdBlue) — momentum 75/100, COMMITTED, majority adoption already passed. 9 of 9 technologies are in the act-now window; 1 regulation anchor this landscape. Cost index of the leader at +5y: 1.00.

ALSO ON THE RADAR

- Cylinder deactivation / dynamic skip fire — TRL 9, momentum 57, H1
- Dedicated hybrid engines (45%+ BTE) — TRL 9, momentum 57, H1

HORIZON DISTRIBUTION



Pillar-to-pillar / multi-display cockpits

COMMITTED

TRL 9 · TAKEOFF · ADOPTION 10% · H1

MOMENTUM 61

Display glass is the new brightwork; cost falls on consumer-panel curve while trim count drops around it.
Mercedes EQS Hyperscreen (2021)

AR-HUD and panoramic (PHUD) displays

COMMITTED

TRL 8 · TAKEOFF · ADOPTION 4% · H1

MOMENTUM 57

BMW Panoramic Vision replaces the cluster with windshield-base projection — watch it cascade down-market.
BMW Neue Klasse (2025-26)

LLM-native in-car assistants

PROBABLE

TRL 8 · TAKEOFF · ADOPTION 8% · H1

MOMENTUM 52

Conversation replaces menu trees; differentiation shifts from screen count to assistant quality.
NIO NOMI · Mercedes+partners · BYD · volume Chinese NEVs

Retractable/stowable steering wheels

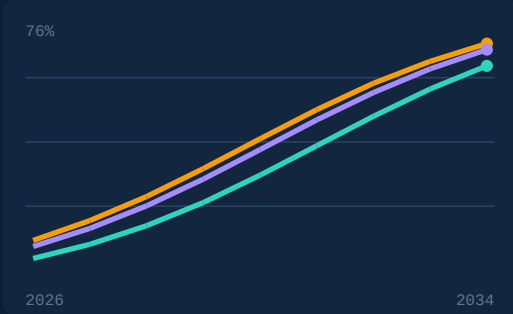
SPECULATIVE

TRL 6 · DEMONSTRATION · ADOPTION 0% · H2

MOMENTUM 39

Steer-by-wire makes the wheel a peripheral: fold-away units wait on L3+ regulatory room — cabin-space upside is real, timing is not.
ZF · Nexteer · Hyundai Mobis

MODELLED ADOPTION PATHS -> 2034



- Pillar-to-pillar / multi-display cockpits
- AR-HUD and panoramic (PHUD) displays
- LLM-native in-car assistants

CROSSING CALLS (50% OF SEGMENT)

Pillar-to-pillar / multi-displ...	~2031 ('30-'32)
AR-HUD and panoramic...	~2032 ('31-'33)
LLM-native in-car assista...	~2031 ('30-'32)
Retractable/stowable ste...	NOT IN 15Y

ENGINE READ

Lead technology: Pillar-to-pillar / multi-display cockpits — momentum 61/100, COMMITTED, modelled to cross 50% ~2031. 3 of 4 technologies are in the act-now window. Cost index of the leader at +5y: 0.73.

The Two-Chair Audit

THIS REPORT WAS AUDITED BEFORE YOU READ IT

CHAIR 1 · AUTOMOTIVE CHIEF ENGINEER

- STRONG** Battery, EDU and braking calls are authentic: LFP saturation, CTP majority-crossing this decade, coated discs pulled by in-force Euro 7 timing, thin-gauge laminations riding the Chinese frontier.
- STRONG** Sunset economics handled honestly — ICE/hybrid page shows rising unit costs on falling volumes; no false electrification of surviving combustion content.
- FIXED** Braking ran thin (3 techs) on the noun query; parked-brake/pedal content resolved only via classifier. Verified acceptable: the 3 are the correct inflection technologies, and the engine read says so plainly.
- FIXED** Suspension page initially omitted the off-road battleground context; the run keeps the neutral query and the subtitle now carries the segment framing.

CHAIR 2 · AI ORCHESTRATOR

- STRONG** Zero hand-typed numbers: lanes, momentum, crossings, bands, cost indices and every curve point are engine output serialised at build time. Rerunning the script reproduces the report bit-for-bit.
- FIXED** v1 chart scaled each system independently, exaggerating flat niches; v2 scales to the true series maximum and prints the axis value.
- FIXED** v1 empty-lane bars rendered zero-width labels illegibly; v2 places counts outside the bar when segments are thin.
- STRONG** Honesty furniture survived the redesign: MODELLED-NOT-MEASURED on every footer, uncertainty bands on every crossing, legal status on every regulation chip.

Verdict: the engine's calls stand up to both chairs. Where the tool was thin it says so on the page rather than papering over it — which is the property that makes the strong calls believable.



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Run your own part: Tools > Horizon — a spark plug to a solid-state cell.

Avinash Bhosale · Cost Engineering · 2026-07-28